Text as Data

W7: Scaling Political Texts

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Text labels and continuous positions

Research task	W6 connection	W7 question
Assign a category	Hate speech and grandstanding	Which texts belong together?
Measure a degree	Nielsen's ideological language score	Where does a text sit on a scale?
Interpret the measure	Read errors and external evidence	What gives a position its meaning?

Today: Wordscores, Wordfish, Wordshoal and Latent Semantic Scaling (LSS).

Course connection: W6: Human Coding and Document Classification. Lecture context: Benoit (2018), pp. 3–4, 21–27.

Learning objectives

- Explain the problem each method addresses and its main assumptions.
- Connect a published application to its texts and measurement target.
- Distinguish reference scores, latent positions and seed words.
- Read the texts and plan independent checks before fitting a model.

Sequence: cases → method concepts
→ reading and validation → R tools and practice.

Research cases and original-study sources follow. Method selection depends on the question and available evidence.

Research cases in W7

Study	Article	Texts	Teaching connection
Study 1	Laver, Benoit & Garry (2003)	Party manifestos	Wordscores and reference positions
Study 2	Slapin & Proksch (2008)	German manifestos	Wordfish and change over time
Study 3	Lauderdale & Herzog (2016)	Legislative debates	Wordshoal and debate structure
Study 4	Trubowitz & Watanabe (2021)	News summaries	LSS and geopolitical threat
Study 5	Hjorth et al. (2015)	Danish and German manifestos	External benchmarks and context
Study 6	Lowe & Benoit (2013)	Irish budget speeches	Human judgments and model failure

Wordscores · Wordfish · Wordshoal · GTI · External benchmarks · Human validation.

Wordscores: reference texts and new documents

Situation: some texts have credible positions on a defined dimension.

- Use these documents as **reference texts**.
- Learn how their words distinguish those positions.
- Estimate positions for comparable new documents.

Example: earlier party manifestos provide references for a later election.

Laver, Benoit & Garry (2003), APSR, pp. 312–314. Lecture context: Benoit (2018), pp. 4, 17–20.

Wordscores: assumptions and scope

- **Reference validity:** assigned scores represent the intended concept.
- **Comparability:** references and new texts use a sufficiently comparable political language.
- **Coverage:** new texts share enough words with the references.
- **Transfer:** the relationship between vocabulary and position remains useful across the comparison.

Discuss: would a campaign manifesto be a suitable reference for a parliamentary interruption?

Laver, Benoit & Garry (2003), selection of reference texts, pp. 314–315. Discussion prompt is a course application. Lecture context: Benoit (2018), pp. 18–20.

Laver, Benoit and Garry: party positions

Study 1

American Political Science Review Vol. 97, No. 2 May 2003

Extracting Policy Positions from Political Texts Using Words as Data

MICHAEL LAVER and KENNETH BENOIT *Trinity College, University of Dublin*
JOHN GARRY *University of Reading*

Michael Laver, Kenneth Benoit and John Garry

American Political Science Review, 2003

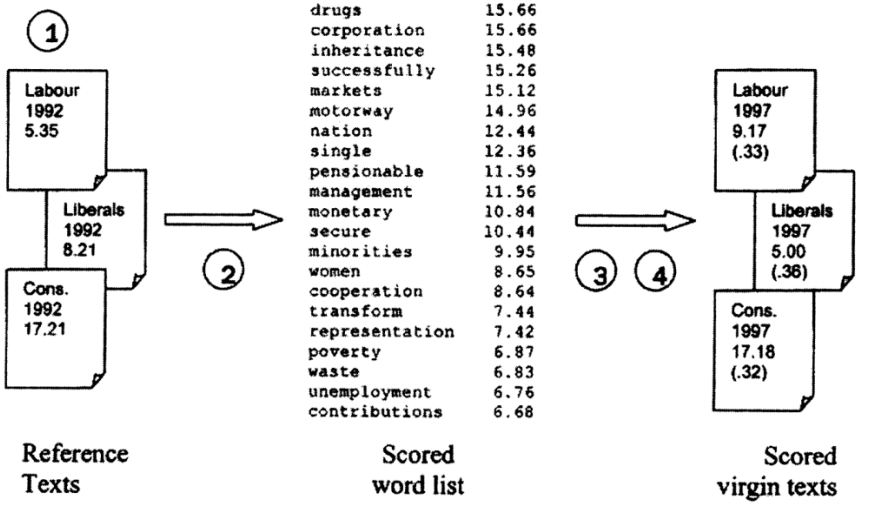
- **Question:** how can texts locate parties on policy dimensions?
- **References:** British 1992 manifestos and expert scores.
- **New texts:** the parties' 1997 manifestos.

Laver, Benoit & Garry (2003), "Extracting Policy Positions from Political Texts Using Words as Data". Lecture walkthrough: Benoit (2018), pp. 5–8.

Wordscores: interpreting estimated change

Study 1

FIGURE 1. The Wordscore procedure, using the British 1992–1997 manifesto scoring as an illustration



Step 1: Obtain reference texts with a priori known positions (`setref`)
Step 2: Generate word scores from reference texts (`wordscore`)
Step 3: Score each virgin text using word scores (`textscore`)
Step 4: (optional) Transform virgin text scores to original metric

Note: Scores for 1997 virgin texts are transformed estimated scores; parenthetical values are standard errors. The scored word list is a sample of the 5,299 total words scored from the three reference texts.

Figure 1 · reference texts, words and new texts

- Scores inherit the reference dimension.
- The reported 1997 values are **transformed**.
- Change needs comparable language over time.

Laver, Benoit & Garry (2003), APSR, Figure 1, p. 314. Original figure, including its transformation note. Lecture walkthrough: Benoit (2018), pp. 5–8, 15.

Wordfish: positions without reference scores

Situation: we have comparable documents but no trusted position labels.

- Estimate a latent dimension from patterns in word counts.
- Account for document length and the baseline frequency of words.
- Interpret the recovered dimension using texts and external evidence.

Example: party manifestos from several elections, or speeches in one budget debate.

Slapin & Proksch (2008), AJPS. Lowe & Benoit (2013), Political Analysis. Lecture context: Benoit (2018), pp. 34–35, 37.

Wordfish: assumptions about word counts

- **Count model:** word counts follow conditionally independent Poisson distributions.
- **One dimension:** a document position helps explain word use after accounting for document and word baselines.
- **Political interpretation:** the dimension must track the concept we care about.

Possible complications: changing topics, speaking roles, genre and historical vocabulary.

Slapin & Proksch (2008), model specification, pp. 708–710. Interpretation also follows Lowe & Benoit (2013). Lecture context: Benoit (2018), pp. 37–38, 42–43.

Slapin and Proksch: German manifestos

Study 2

A Scaling Model for Estimating Time-Series Party Positions from Texts

Jonathan B. Slapin Trinity College, Dublin
Sven-Oliver Proksch University of California, Los Angeles

Jonathan B. Slapin and Sven-Oliver Proksch

American Journal of Political Science, 2008

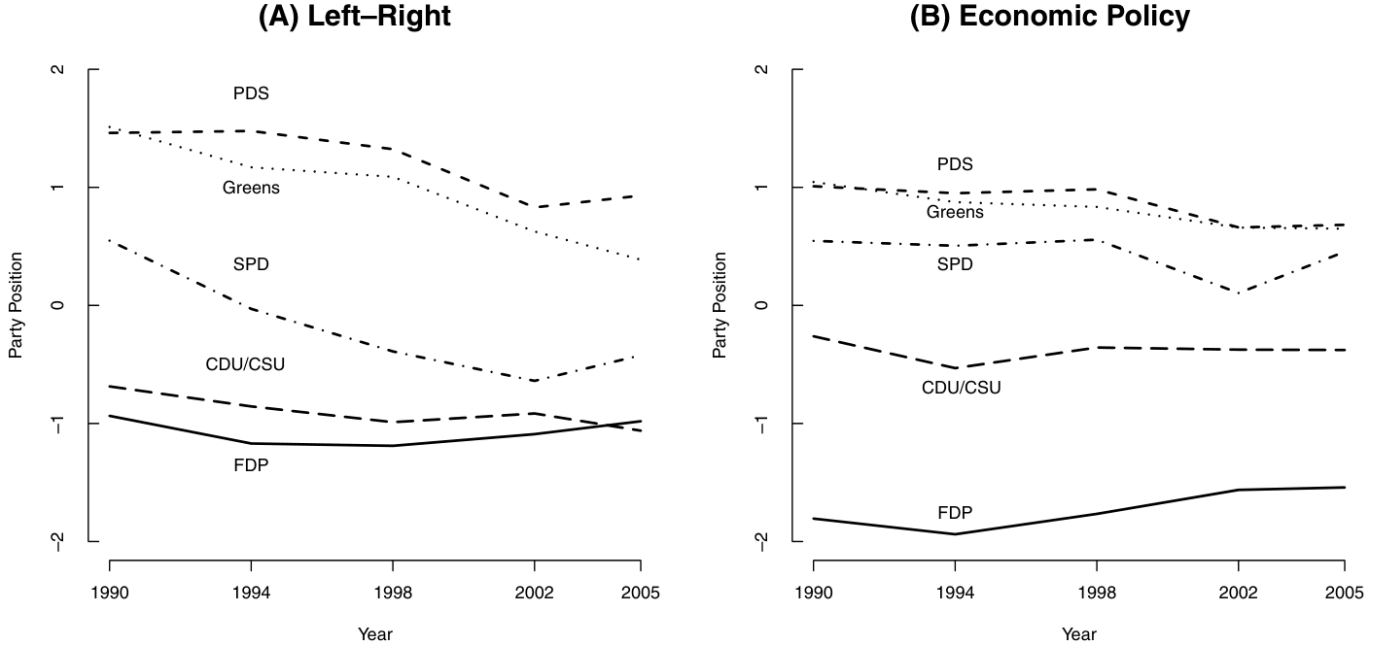
- **Question:** how do party positions change over time?
- **Texts:** German manifestos, 1990–2005.
- **Evidence:** estimated positions and politically discriminating words.

Slapin & Proksch (2008), "A Scaling Model for Estimating Time-Series Party Positions from Texts". Lecture example: Benoit (2018), p. 46.

German party positions across elections

Study 2

FIGURE 1 Estimated Party Positions in Germany, 1990–2005



- Compare parties **within each panel.**
- Panels use different policy scopes.
- A common time scale requires comparable word use.

Figure 1 · panels A and B

Slapin & Proksch (2008), AJPS, Figure 1, p. 714. Original panels A–B, cropped from the four-panel figure. Lecture example: Benoit (2018), p. 46.

Wordshoal: speakers across debates

Situation: legislators speak about different issues in different debates.

- Compare speakers **within each debate** first.
- Estimate a Wordfish position for each speaker in that debate.
- Connect debate-specific positions through legislators who appear across debates.

Target: a common summary of expressed positions across debates.

Lauderdale & Herzog (2016), Political Analysis, §§2–3. Lecture context: Benoit (2018), pp. 47–49.

Wordshoal: assumptions across debates

- **Within debates:** word-use differences provide information about relative positions.
- **Across debates:** some position differences reflect a common dimension.
- **Overlap:** repeated participants connect the debates.
- **Participation:** selective speaking limits what observed positions represent.

The common scale summarizes positions expressed in the debates legislators joined.

Lauderdale & Herzog (2016), Political Analysis, §§2–3 and 6. The second-stage model treats missing debate positions as missing at random. Lecture context: Benoit (2018), pp. 47–49.

Lauderdale and Herzog: legislative speech

Study 3

Measuring Political Positions from Legislative Speech

Benjamin E. Lauderdale
Department of Methodology, London School of Economics, Houghton Street, London WC2A 2AE, UK
e-mail: b.e.lauderdale@lse.ac.uk (corresponding author)

Alexander Herzog
School of Computing, Clemson University, Clemson, SC 29634, USA
e-mail: aherzog@clemson.edu

Benjamin E. Lauderdale and Alexander Herzog

Political Analysis, 2016

- **Question:** how can debates yield legislator-specific positions?
- **Ireland:** 29th and 30th Dáil.
- **United States:** 104th–113th Senates.

Lauderdale & Herzog (2016), "Measuring Political Positions from Legislative Speech", §§3–5. Lecture example: Benoit (2018), pp. 47–50.

Government and opposition in the Irish Dáil

Study 3

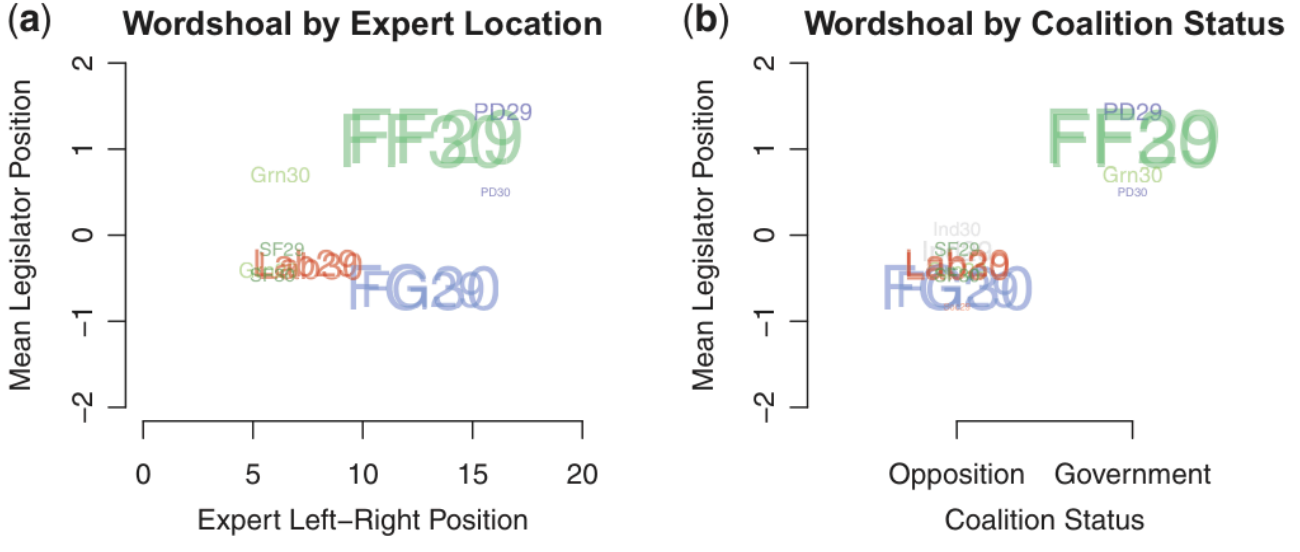


Figure 1 · panels a and b

- Compare two possible interpretations.
- The main dimension follows **coalition status**.
- Left–right needs its own evidence.

Lauderdale & Herzog (2016), Political Analysis, Figure 1(a–b), advance article p. 8. Original panels cropped from the full figure. Lecture example: Benoit (2018), p. 50.

Debate structure and research scope

Study 3

Speaker	Budget	Health	Foreign policy
A	Speaks	Speaks	—
B	Speaks	—	Speaks
C	—	Speaks	Speaks

Teaching example: shared speakers link different debates.

A single budget debate provides no cross-debate structure. A disconnected set of speakers requires additional identifying information.

Illustrative participation table created for this course. Method connection: Lauderdale & Herzog (2016), §3. Lecture context: Benoit (2018), pp. 47–49.

LSS: seed words and a defined concept

Situation: we can name a concept's poles but have few labelled documents.

- Provide words that express each end of the intended dimension.
- Learn semantic relationships from a relevant corpus.
- Use proximity to the seeds to score words, then documents.

Hostility (+1): *enemy, foe, hostile*.
Friendliness (−1): *ally, friend, peaceful*.

Watanabe (2021), "Latent Semantic Scaling". Seed examples from Trubowitz & Watanabe (2021), ISQ. Plural forms and other seeds omitted here.

LSS: assumptions about meaning and context

- **Seeds:** words express the intended poles in this domain.
- **Semantics:** co-occurrence patterns contain useful information about that distinction.
- **Aggregation:** word scores provide a meaningful summary at the document level.
- **Target:** polarity relates to the actor or issue we intend to measure.

Discuss: could a negative news story describe an ally sympathetically?

Watanabe (2021), CMM. LSX: seed selection and target-specific example. Discussion is a course application.

Trubowitz and Watanabe: geopolitical threat

Study 4



The Geopolitical Threat Index: A Text-Based Computational Approach to Identifying Foreign Threats

PETER TRUBOWITZ
London School of Economics and Political Science, UK

AND

KOHEI WATANABE
Waseda University, Japan

Peter Trubowitz and Kohei Watanabe

International Studies Quarterly, 2021

- **Texts:** New York Times summaries, 1861–2017.
- **Models:** Newsmap and LSS.
- **Target:** reported hostile foreign military behavior.

Trubowitz & Watanabe (2021), "The Geopolitical Threat Index". LSS method: Watanabe (2021). Image: published advance article from LSE Research Online.

Hostility words and the threat index

Study 4

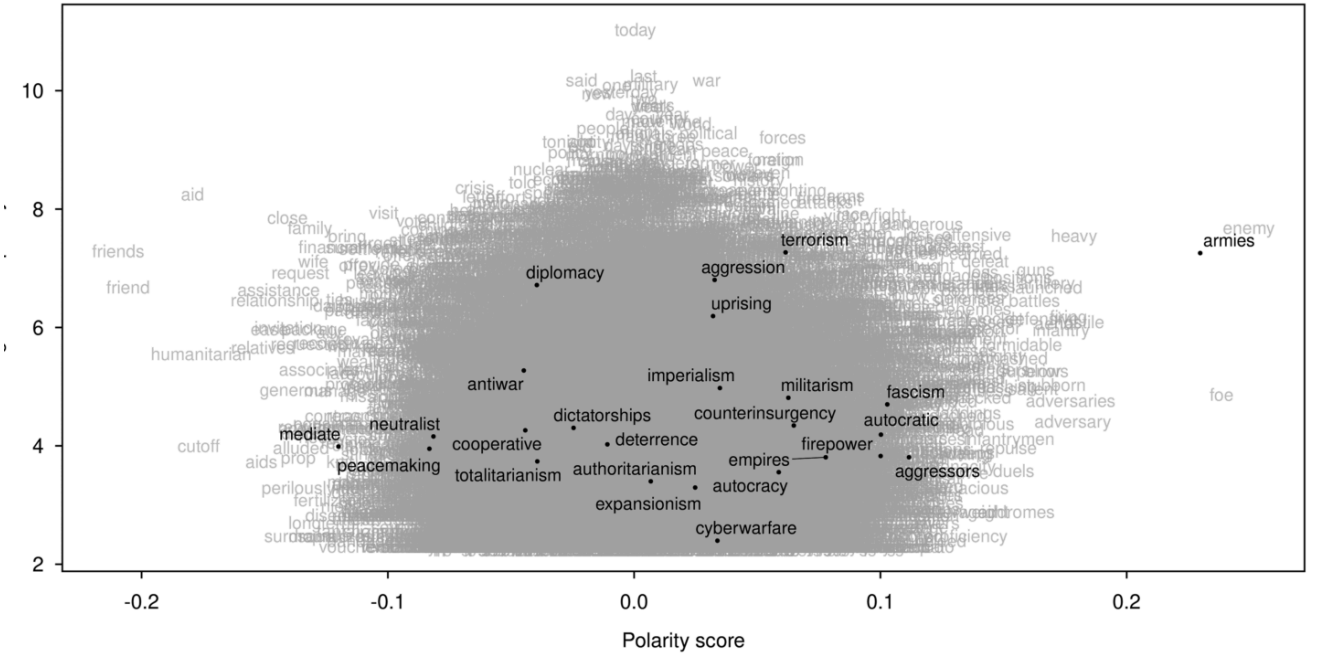


Figure 2 · learned word polarity

- Seeds specify **hostility**.
- Countries and relevant stories require separate identification.
- GTI aggregates the share of hostile military stories.

Trubowitz & Watanabe (2021), ISQ, Figure 2 and "Geopolitical Threat Index." GTI is a proportion, distinct from a mean polarity score.

Validation depends on the level of analysis

Study 4

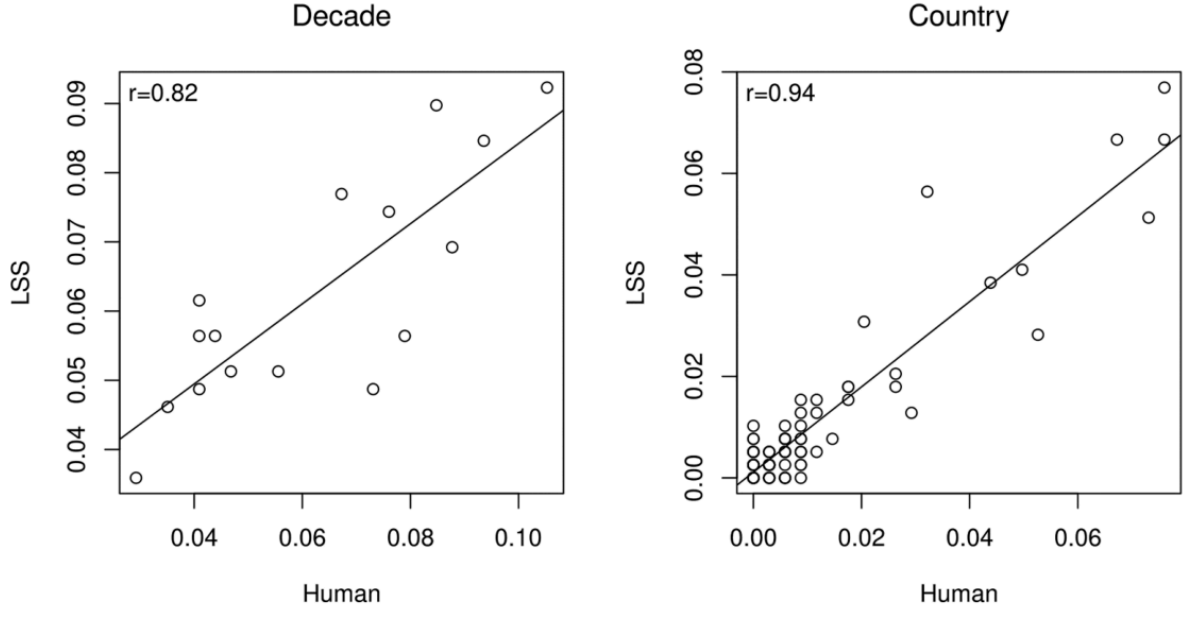


Figure 4 · aggregate hostile-story proportions

- By decade: **r = 0.82**.
- By country: **r = 0.94**.
- Individual summaries: about **59% agreement**.

Trubowitz & Watanabe (2021), ISQ, Figure 4 and footnote 18. Correlations describe aggregate proportions, not individual-story accuracy.

Supervision, units and outputs

Studies 1–4

Method	Guidance supplied	Structure	Main output
Wordscores	Reference document scores	Comparable reference and new texts	New document positions
Wordfish	No position labels	One count matrix	Relative document positions
Wordshoal	No position labels	Speakers across debates	Common speaker positions
LSS	Seed words and polarity	Corpus semantic relationships	Word and document polarity

Next: how does each method turn its inputs into positions?

Laver et al. (2003) · Slapin & Proksch (2008) · Lauderdale & Herzog (2016) · Watanabe (2021). Lecture context: Benoit (2018), pp. 4, 34–35, 47–49.

Wordscores and Naive Bayes

W6 connection: Nielsen already used reference language to construct a continuous measure.

Method	What it combines	Output
Naive Bayes classification	Word likelihoods and a class prior	A document's class probability
Linear Wordscores	Word-level weights and reference positions	An average position for the document

A class probability does not automatically measure the degree of a political trait.

Discuss: if every word is repeated twice, should the estimated political position change?

Benoit (2018), pp. 3–4, 21–27; Laver, Benoit & Garry (2003). The comparison concerns a fixed fitted model.

Wordscores 1: normalize the references

Constructed example: reference A = −1; reference B = +1. No smoothing.

Word	A count	B count	A share	B share
spending	40	20	0.40	0.10
cuts	10	80	0.10	0.40
budget	50	100	0.50	0.50
Total	**100**	**200**	**1.00**	**1.00**

$$f_{rj} = \text{count of word j in reference r / total words in r}$$

Use word shares so that document length alone does not give B more influence.

Invented counts for teaching. Benoit (2018), p. 9; Laver, Benoit & Garry (2003). A and B are illustrative reference documents.

Wordscores 2: give each word a score

Normalize each word's shares across the references, using equal prior weight.

Word	A weight	B weight	Word score
spending	0.80	0.20	−0.60
cuts	0.20	0.80	+0.60
budget	0.50	0.50	0.00

$$p_{rj} = f_{rj} / \sum_{r'} f_{r'j}$$

$$s_j = \sum_r p_{rj} a_r$$

For spending: **0.80 × (−1) + 0.20 × (+1) = −0.60.**

Constructed example continued. Benoit (2018), pp. 10–13, 21–25; Laver, Benoit & Garry (2003). a_r is the assigned reference position.

Wordscores 3: score a new document

A new text contains **50 tokens**, of which **40 have learned scores**.

Word	Count	Share of scored tokens	Contribution
spending	10	0.25	−0.15
cuts	20	0.50	+0.30
budget	10	0.25	0.00
jobs	10	Excluded: no reference score	—

$$v = 0.25(-0.60) + 0.50(+0.60) + 0.25(0) = +0.15$$

Coverage: 40 / 50 = 80%. What information could the unscored words contain?

Constructed example continued. Benoit (2018), p. 14; Laver, Benoit & Garry (2003); quanteda.textmodels manual. The illustration matches to the scored vocabulary first.

Why raw Wordscores move toward the centre

Half of each reference is **budget**, whose learned score is zero.

Reference	Assigned position	Raw score when rescored
A	−1.00	−0.18
B	+1.00	+0.18

$$\text{A: } 0.40(-0.60) + 0.10(+0.60) + 0.50(0) = -0.18$$

Rescoring a reference need not reproduce its assigned position.

Practice choice: use raw scores (`rescaling = "none"`) and report that choice.

Constructed example continued. Benoit (2018), pp. 14–15; Laver, Benoit & Garry (2003); quanteda.textmodels manual. Rescaling choices do not establish validity.

Wordfish: explain expected word counts

$$y_{ij} \sim \text{Poisson}(\mu_{ij})$$
$$\log \mu_{ij} = \alpha_i + \psi_j + \beta_j \theta_i$$

Parameter	Role in the expected count
α_i	Document baseline, related to length
ψ_j	Word's baseline frequency
θ_i	Document's relative position
β_j	How that word's use changes with position

The exponential link makes expected counts positive. Conditional Poisson variance equals its mean.

Benoit (2018), pp. 37–38, 42; Slapin & Proksch (2008). The statistical model and its political interpretation are separate claims.

Wordfish: what does a word loading do?

Constructed example: hold the baseline at $\alpha_i + \psi_j = \log 2$.

Word loading	μ when $\theta = -1$	μ when $\theta = 0$	μ when $\theta = +1$
$\beta = -1$	5.44	2.00	0.74
$\beta = 0$	2.00	2.00	2.00
$\beta = +1$	0.74	2.00	5.44

$$\mu = 2 \times \exp(\beta\theta)$$

Discuss: which parameter describes a word's baseline frequency, and which describes separation?

Expected counts calculated from an illustrative parameter setting, not a fitted dataset. Benoit (2018), pp. 37–38; Slapin & Proksch (2008).

Wordfish: coordinates need identification

Several parameter descriptions can produce the same expected counts.

Change to the positions	Compensating change
Shift every θ by the same amount	Adjust word baselines
Multiply every θ by a positive constant	Divide β by that constant
Reverse every θ	Reverse every β

$$\beta_j \theta_i = (-\beta_j)(-\theta_i)$$

`dir = c(a, b)` chooses $\theta_a < \theta_\beta$. It does not assign scores or define a policy dimension.

Benoit (2018), p. 41; Slapin & Proksch (2008); quanteda.textmodels manual. The current package interface supplies the ordering convention.

Wordfish: alternate the parameter updates

1. **Initialize** document positions and word parameters.
2. **Hold words fixed:** update document baselines and positions, α and θ.
3. **Hold documents fixed:** update word baselines and loadings, ψ and β.
4. **Normalize and repeat** until the numerical convergence criteria are met.

Each conditional update resembles a Poisson regression with some parameters held fixed.

Discuss: what does convergence tell us about the intended political interpretation?

Benoit (2018), pp. 39–40; quanteda.textmodels manual. This is the conditional-update intuition; implementation also includes its documented priors and tolerances.

Wordfish: where the count model can fail

Model condition	Textual complication
Conditional independence	Collocations, quotations and repeated phrases
Poisson variance = mean	Informative words occurring in bursts
One dimension captures the target	Topics, speaking roles or several political conflicts

Correlated words can provide less independent information than the model assumes.

Discuss: could a narrow model interval surround a position that measures the wrong concept?

Benoit (2018), pp. 42–45, 60–61; Lowe & Benoit (2013). The table describes possible failures, not a diagnosis of every corpus.

Wordshoal: linking debate positions

First stage: estimate a Wordfish position $z_{i\,j}$ within each debate j.

$$z_{ij} = a_j + b_j\theta_i + \varepsilon_{ij}$$

Quantity	Meaning
θ_i	Speaker's common position
b_j	Direction and strength of a debate's connection
Missing $z_{i\,j}$	The legislator did not speak in that debate

Debate scales can connect with different signs and strengths.

Teaching notation for Lauderdale & Herzog (2016), §3.2. The original second-stage model also specifies speaker precision and priors. Lecture context: Benoit (2018), pp. 47–49.

Wordshoal: align debate scales first

Constructed example: two debates encode the same ordering with opposite signs.

Speaker	Debate A position	Debate B position	Naive average
A	−1	+1	0
B	0	0	0
C	+1	−1	0

A common model must align debate locations, scales and directions before interpreting shared positions.

Discuss: why does a negative debate loading not necessarily indicate an absence of common structure?

Invented positions for teaching. Benoit (2018), pp. 47–49; Lauderdale & Herzog (2016). This is not output from a Wordshoal fit.

LSS: semantic information and aggregation

Step	Information used
Define poles	Concept-specific seed words and polarity
Estimate semantic relationships	Word co-occurrence, typically across sentences
Score words	Similarity to the seed words in a reduced semantic space
Score documents	Frequency-weighted average of word scores

Units matter: sentence-level learning can support article-level scoring and later country-year aggregation.

Watanabe (2021), CMM. LSX workflow and grouping. Country-year aggregation refers to the GTI application. Vector background: Benoit (2018), pp. 68–69.

LSS: from vectors to word polarity

Use singular value decomposition (**SVD**) to reduce sentence–word counts to **k dimensions**.

$$g_j = (1 / |S|) \sum_{s \in S} \cos(v_s, v_j)\, p_s$$

Quantity	Meaning
v_j	Word vector learned from sentence contexts
p_s	Researcher-assigned polarity of seed s, such as −1 or +1
g_j	Word polarity: similarity weighted by the seed polarities

Document scores then average word polarities using token frequencies.

Watanabe (2021), "Latent Semantic Scaling," Eqs. 1–2. Representation background: Benoit (2018), pp. 54, 68–69.

From W6 evaluation to W7 validation

Question	W6 classification practice	W7 scaling
What is evaluated?	Predicted categories	Positions on a stated dimension
What is the evidence?	Reserved human labels; F1	External measures; rank agreement
What do we read?	Errors and coding boundaries	Surprising positions and their language

Hjorth et al. test convergence across measures. This differs from k-fold cross-validation.

Next: do Wordscores and Wordfish agree with external measures of party positions?

Course synthesis connecting W6, Hjorth et al. (2015) and Lowe & Benoit (2013). Validation framework: Benoit (2018), p. 59.

Hjorth et al.: validating party positions

Study 5

Computers, coders, and voters: Comparing automated methods for estimating party positions

Frederik Hjorth[1],Robert Klemmensen[2],Sara Hobolt[3], Martin Ejnar Hansen[4],Peter Kurrild-Klitgaard[5]

Frederik Hjorth, Robert Klemmensen, Sara Hobolt, Martin Ejnar Hansen and Peter Kurrild-Klitgaard

Research & Politics, 2015

- **Question:** do text scales agree with other measures of party positions?
- **Comparison:** Wordscores and Wordfish.
- **Scope:** 254 manifestos across 33 elections.

Hjorth et al. (2015), "Computers, Coders, and Voters", pp. 1–2. Original article heading, CC BY-NC 3.0.

Two countries, three validation benchmarks

Study 5

Country	Elections	Manifestos	Mean words
Denmark, 1945–2007	24	212	1,232
Germany, 1980–2009	9	42	10,306

- **Experts:** published assessments of party positions.
- **Voters:** mean left–right self-placement of each party's supporters.
- **CMP RILE:** a left–right index from human-coded manifestos.

Assumption to examine: do these benchmarks address the same political dimension?

Hjorth et al. (2015), pp. 2–4 and Table 1. Mean word counts rounded. CMP = Comparative Manifesto Project.

Validation differs across the two countries

Study 5

Original Table 3: mean Spearman rank correlations across elections.

Country	Method	CMP	Experts	Voters
Denmark	Wordscores	0.7	0.8	0.8
Denmark	Wordfish	0.2	0.2	0.1
Germany	Wordscores	0.8	0.8	0.8
Germany	Wordfish	0.8	0.7	0.7

Rank agreement supports an ordering; it does not establish meaningful distances.

Discuss: why might Wordfish agree with the benchmarks in one country more than the other?

Hjorth et al. (2015), Table 3, p. 7. Published correlations transcribed; estimation times omitted. These are not classification accuracies.

Does text length explain the difference?

Study 5

1. **Check:** randomly shorten German manifestos to lengths comparable to the Danish texts.
2. **Finding:** part of the German Wordfish advantage disappears; part remains.
3. **Interpretation:** length matters; ideological vocabulary is another possible contributor.

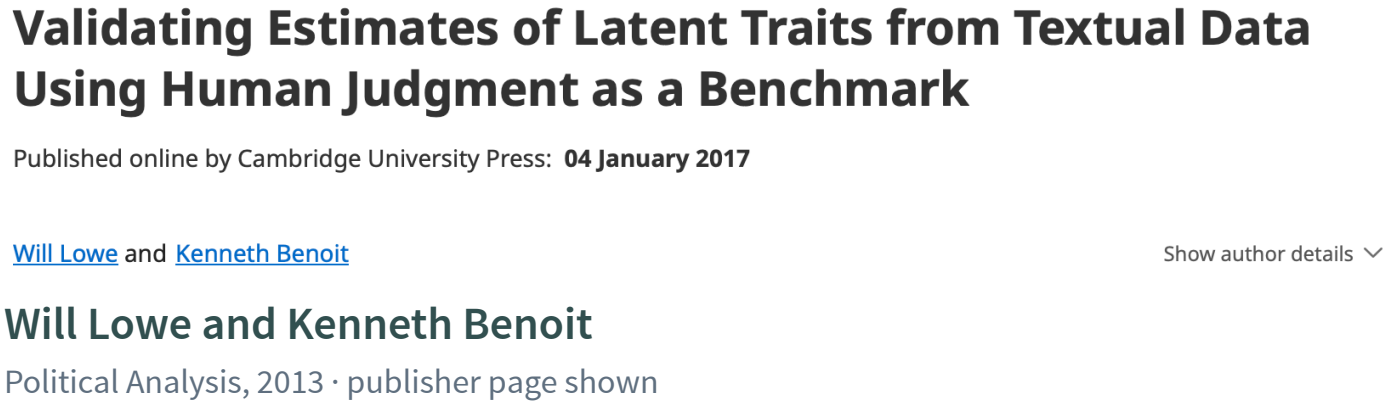
The two-country comparison does not establish a universal minimum text length.

Discuss: would combining more speeches add useful information, or mix topics and speaking roles?

Hjorth et al. (2015), "Difference in cross-validity" and discussion, pp. 4–6. Shortening results summarized by the authors from their online appendix.

Lowe and Benoit: the Irish budget debate

Study 6



Validating Estimates of Latent Traits from Textual Data Using Human Judgment as a Benchmark

Published online by Cambridge University Press: **04 January 2017**

Will Lowe and Kenneth Benoit

Show author details

Will Lowe and Kenneth Benoit

Political Analysis, 2013 · publisher page shown

- **Texts:** 14 speeches on the 2010 Irish budget.
- **Debate:** December 2009.
- **Benchmark:** human judgments of budget support.

Lowe & Benoit (2013), Political Analysis, §3. Image: publisher page (user-provided screenshot).

Human judgments expose a model failure

Study 6

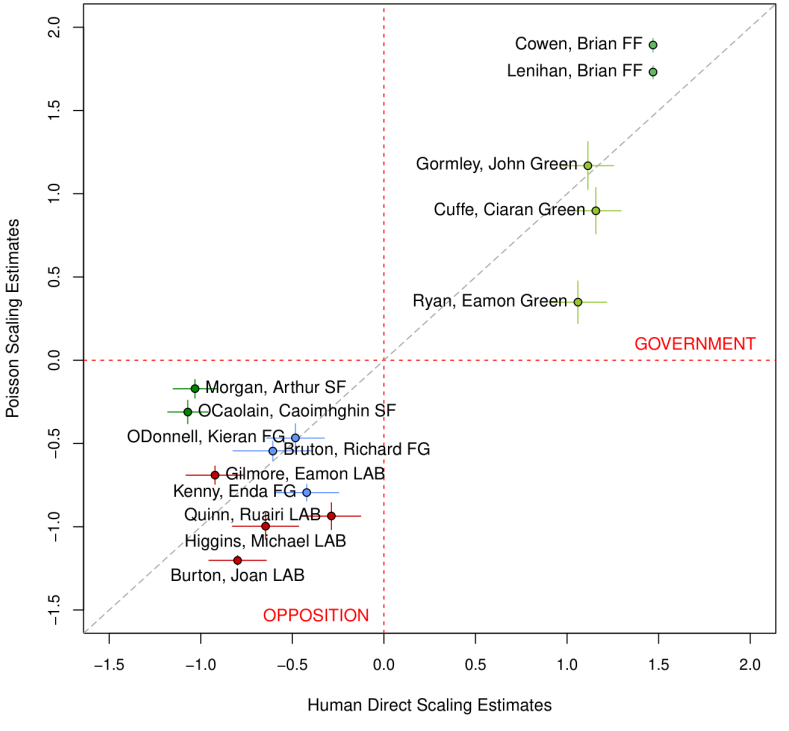


Figure 3: *Direct comparison of Human v. Wordfish Estimates.*

Figure 3 · human placement and Wordfish

- Many positions broadly agree.
- **Sinn Féin:** strong opposition looks too moderate in Wordfish.
- A second political distinction complicates one dimension.

Lowe & Benoit (2013), Political Analysis, §§5–6. Figure 3 in the author manuscript, p. 32. Dimensionality background: Benoit (2018), p. 63.

Reading budget speeches without party labels

Study 6 · Reading exercise

Text A

> We have sought to do all of this in a manner that is fair and that protects the most vulnerable.

Text B

> This is a jobless and a joyless budget.

Discuss: which passage supports the budget? What else would you read before assigning a position?

Verbatim excerpts from `quanteda.textmodels::data_corpus_irishbudget2010`. Speaker identities follow on the next slide. Corpus documentation. Related reading example: Benoit (2018), pp. 64–66.

Budget support and speaking roles

Study 6 · Reading exercise

> We have sought to do all of this in a manner that is fair and that protects the most vulnerable.

Brian Lenihan · Fianna Fáil · Minister for Finance

> This is a jobless and a joyless budget.

Richard Bruton · Fine Gael · opposition finance spokesperson

Interpretation: budget support, policy preferences and government responsibility can overlap in these speeches.

Irish 2010 budget corpus, Lenihan sentence 7 and Bruton sentence 6 under quanteda 4.3.1 sentence segmentation. Context: Lowe & Benoit (2013), §3. Related role example: Benoit (2018), pp. 64–66.

Gormley: one speaker, two roles

Study 6 · Lecture example

Benoit's fuller-speech example: Green Party leader and government minister.

> "As leader of the Green Party …"

The speaker sets out his party's budget position.

> "I will now comment on some specific aspects of my Department's Estimate."

The speaker turns to departmental business.

Discuss: which passages belong in a measure of budget support?

Benoit (2018), pp. 64–66; short excerpts from p. 65. The lecture's text scope differs from our classroom corpus.

Quoted praise in an opposition speech

Study 6 · Reading exercise

Enda Kenny’s speech includes:

> an honest and credible attempt to meet the country’s need

The passage refers to an **Irish Independent editorial about the April 2009 supplementary budget**.

Kenny then says that the earlier expectations had failed.

Discuss: whose evaluation would a word-based score capture here?

Irish budget corpus, Kenny sentences 3–11 under quanteda 4.3.1. The short quotation is an excerpt from the editorial passage reproduced in his speech. Document: “Kenny, Enda (FG).” Representation background: Benoit (2018), p. 68.

Anchors and independent validation

Evidence	What it does	What it leaves to check
Wordscores references	Define the learned scale	Performance on other texts
Wordfish `dir`	Choose an orientation	Meaning of the dimension
LSS seeds	Specify a semantic direction	Concept and target in context
Independent judgments	Check an intended interpretation	Agreement, coverage and scope

Party separation and agreement between models provide partial evidence.

Course synthesis of Lowe & Benoit (2013) and the model documentation. Judgments used to tune a measure should be separated from its final evaluation. Validation framework: Benoit (2018), p. 59.

Four questions for validating a scale

Evidence	Question to investigate
Semantic validity	Do extreme texts and discriminating words express the intended concept?
Convergent / discriminant validity	Does the scale agree with relevant measures and differ from competing constructs?
Predictive validity	Does it support a specified prediction on new observations?
Hypothesis validity	Are its relationships consistent with substantive theory?

A theoretically expected result is one piece of evidence. Keep final checks separate from model tuning.

Teaching adaptation of Benoit (2018), p. 59; Lowe & Benoit (2013) and Hjorth et al. (2015).

Robustness and uncertainty

Sensitivity: replace references or seeds, change the text scope, and compare reasonable preprocessing choices.

Uncertainty: distinguish model intervals, human disagreement and variation across specifications.

Evaluation: reserve judgments that did not guide those choices.

Discuss: which choice would you vary first, and what evidence would you reserve?

Lowe & Benoit (2013), uncertainty and model assumptions. Watanabe, seed-word evaluation. Evaluation design is a course synthesis. Lecture context: Benoit (2018), pp. 60–66.

A validation plan for your research abstract

1. **Concept:** what does a high score mean?
2. **Unit:** which text, speaker or period receives the score?
3. **Design:** what information and structure does the method require?
4. **Assumption:** which condition is most likely to fail in your data?
5. **Evidence:** what independent observation could reveal that failure?

Research title and abstract: due **October 23, 2026, before 23:59**.

Course schedule. Deadline follows the W6 announcement. Validation questions synthesize the W7 research cases. Validation framework: Benoit (2018), p. 59.

R tools for text scaling

Task	Package	Main interface
Prepare texts and counts	`quanteda`	Corpus, tokens and DFM
Reference-based scaling	`quanteda.textmodels`	`textmodel_wordscores()`
Latent count scaling	`quanteda.textmodels`	`textmodel_wordfish()`
Cross-debate scaling	`wordshoal`	`textmodel_wordshoal()`
Semantic scaling	LSX	`textmodel_lss()`

R practice: Wordscores and Wordfish on one budget debate.

quanteda · quanteda.textmodels · wordshoal · LSX.

R practice: one debate, two models

Target: support for the 2010 Irish budget.

- Fit Wordscores and Wordfish to the same fourteen speeches.
- Compare rankings and read the largest disagreement.
- Replace one reference and assess sensitivity.

Work in pairs. Run S01–S08b, including lettered substeps, in one R session.
Lab HTML · Lab ZIP.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 59, 64–66.

S01: Packages and the Irish corpus

Copy and run S01.

```r
# S01: Packages and the built-in corpus
# install.packages(c("quanteda", "quanteda.textmodels"))
library(quanteda)
library(quanteda.textmodels)
quanteda_options(threads = 1)
options(width = 72, digits = 4)
corp_irish <- data_corpus_irishbudget2010
c(quanteda = as.character(packageVersion("quanteda")),
  textmodels = as.character(packageVersion("quanteda.textmodels")))
ndoc(corp_irish)
```

The corpus is included in quanteda.textmodels. Keep this R session open for all later steps.

S01 output: packages and documents

R console output

```
  quanteda textmodels
   "4.3.1"   "0.9.10"
[1] 14
```

Discuss: What does one document in this corpus represent?

S02a: Speakers and document IDs

Copy and run S02a.

```r
# S02a: Speakers and stable document IDs
meta <- data.frame(
  doc_id = paste0("ie_", docvars(corp_irish, "number")),
  speaker = docvars(corp_irish, "name"),
  party = docvars(corp_irish, "party"))
docnames(corp_irish) <- meta$doc_id
print(meta, row.names = FALSE)
```

Each row is one speech. The stable IDs keep texts, references and model results aligned.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP.

S02a output: the fourteen speakers

R console output

```
  doc_id  speaker party
   ie_01  Lenihan    FF
   ie_02   Bruton    FG
   ie_03   Burton   LAB
   ie_04   Morgan    SF
   ie_05    Cowen    FF
   ie_06    Kenny    FG
   ie_07 ODonnell    FG
   ie_08  Gilmore   LAB
   ie_09  Higgins   LAB
   ie_10    Quinn   LAB
   ie_11  Gormley Green
   ie_12     Ryan Green
   ie_13    Cuffe Green
   ie_14 OCaolain    SF
```

Discuss: Which information identifies a text, and which information could help interpret its position?

S02b: Tokenize a working copy

Copy and run S02b.

```r
# S02b: Tokenize a working copy
toks_irish <- tokens(corp_irish, remove_punct = TRUE,
  remove_symbols = TRUE, remove_numbers = TRUE)
toks_irish <- tokens_tolower(toks_irish)
head(as.list(toks_irish)[[1]], 12)
```

Remove punctuation, symbols and numbers, then lowercase the working tokens. Keep the original corpus for reading.

S02b output: words from the first speech

R console output

```
 [1] "when"          "i"             "presented"     "the"
 [5] "supplementary" "budget"        "to"            "this"
 [9] "house"         "last"          "april"         "i"
```

Discuss: What information about quotation or emphasis has this representation removed?

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP.

S02c: Build the count matrix

Copy and run S02c.

```
# S02c: Build the common count matrix
counts <- dfm(toks_irish)
dim(counts)
any(ntoken(counts) == 0)
topfeatures(counts, 6)
```

Both methods start from these token counts. No TF-IDF weighting is applied.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 37–38.

S02c output: counts and common words

R console output

```
[1]   14 4885
[1] FALSE
 the   to   of  and   in    a
3598 1633 1537 1359 1231 1013
```

Discuss: What do the rows and columns represent, and why might a frequent word still have little discriminating value?

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 37–38.

S03: Define the Wordscores references

Copy and run S03.

```
# S03: Define the Wordscores references
ref_scores <- rep(NA_real_, ndoc(counts))
names(ref_scores) <- docnames(counts)
ref_ids <- c("ie_01", "ie_02")
ref_scores[ref_ids] <- c(1, -1)
print(data.frame(speaker = meta$speaker[match(ref_ids, meta$doc_id)],
  reference = ref_scores[ref_ids]), row.names = FALSE)
```

Assign +1 to Lenihan and −1 to Bruton for budget support. Leave the other twelve positions unknown.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 17–20.

S03 output: the reference design

R console output

```
  speaker reference
  Lenihan         1
   Bruton        -1
```

Discuss: What textual evidence supports these reference choices, and what does NA mean here?

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 17–20.

S04a: Fit Wordscores and read word scores

Copy and run S04a.

```
# S04a: Fit Wordscores and inspect word scores
ws <- textmodel_wordscores(counts, y = ref_scores, smooth = 0)
probe_words <- c("budget", "government", "tax",
                 "jobs", "fair", "recovery")
print(data.frame(word = probe_words,
  score = round(coef(ws)[probe_words], 3)), row.names = FALSE)
```

With smooth = 0, words must occur in the reference vocabulary to receive a score.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 9–13.

S04a output: six learned word scores

R console output

```
       word  score
     budget -0.384
 government  0.140
        tax  0.665
       jobs  0.379
       fair -0.229
   recovery  0.403
```

Discuss: Why can fair receive a negative score on a budget-support scale?

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 9–13.

S04b: Check the scoring vocabulary

Copy and run S04b.

```
# S04b: Align the scoring vocabulary and inspect coverage
ws_counts <- dfm_match(counts, features = names(coef(ws)))
ws_coverage <- ntoken(ws_counts) / ntoken(counts)
dim(ws_counts)
summary(ws_coverage)
```

Match the DFM to words scored by the references. Coverage is the share of each speech's processed tokens that remain.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 14, 19.

S04b output: vocabulary and token coverage

R console output

```
[1]   14 2131
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.836   0.850   0.874   0.886   0.887   1.000
```

Discuss: Does an 84% coverage rate mean 84% accuracy? What does it actually count?

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 14, 19.

S04c: Predict raw Wordscores

Copy and run S04c.

```r
# S04c: Predict raw Wordscores for every speech
ws_pred <- predict(ws, newdata = ws_counts, rescaling = "none")
results <- meta
results$reference <- ref_scores
results$ws_raw <- as.numeric(ws_pred)
results$coverage <- ws_coverage
print(data.frame(speaker = results$speaker, party = results$party,
  reference = results$reference,
  ws_raw = round(results$ws_raw, 3)), row.names = FALSE)
```

The reference column records assigned inputs. ws_raw records predictions, including reference-text rescoring.

S04c output: reference and new speech scores

R console output

```
   speaker party reference ws_raw
   Lenihan    FF         1  0.339
    Bruton    FG        -1 -0.339
    Burton   LAB        NA -0.001
    Morgan    SF        NA  0.023
     Cowen    FF        NA  0.180
     Kenny    FG        NA -0.020
  ODonnell    FG        NA -0.029
   Gilmore   LAB        NA -0.012
   Higgins   LAB        NA -0.040
     Quinn   LAB        NA -0.023
   Gormley Green        NA  0.159
      Ryan Green        NA -0.012
     Cuffe Green        NA  0.003
  OCaolain    SF        NA  0.026
```

Discuss: Why are the reference texts' predicted scores closer to zero than their assigned values of −1 and +1?

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 14–15.

S05a: Fit Wordfish and set its direction

Copy and run S05a.

```r
# S05a: Fit Wordfish and fix its orientation
wf_dir <- match(c("ie_02", "ie_01"), docnames(counts))
set.seed(7197)
wf <- textmodel_wordfish(counts, dir = wf_dir,
  dispersion = "poisson")
data.frame(speaker = meta$speaker[wf_dir],
  theta = round(wf$theta[wf_dir], 3))
```

Estimate all positions jointly. dir asks for Bruton below Lenihan, using document indices found from stable IDs.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 39–41.

S05a output: the orientation check

R console output

```
  speaker  theta
1  Bruton -0.585
2 Lenihan  1.775
```

Discuss: Does setting the direction teach Wordfish what budget support means?

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 39–41.

S05b: Read Wordfish positions

Copy and run S05b.

```r
# S05b: Inspect Wordfish positions and model standard errors
results$wf_theta <- wf$theta
results$wf_se <- wf$se.theta
print(data.frame(speaker = results$speaker, party = results$party,
  theta = round(results$wf_theta, 3),
  se = round(results$wf_se, 3)), row.names = FALSE)
```

theta gives relative positions. se reports uncertainty under this fitted model and its assumptions.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 59–61.

S05b output: positions and standard errors

R console output

```
   speaker party  theta    se
   Lenihan    FF  1.775 0.021
    Bruton    FG -0.585 0.029
    Burton   LAB -1.150 0.016
    Morgan    SF -0.093 0.029
     Cowen    FF  1.794 0.024
     Kenny    FG -0.789 0.026
  ODonnell    FG -0.493 0.044
   Gilmore   LAB -0.587 0.030
   Higgins   LAB -0.979 0.041
     Quinn   LAB -0.921 0.042
   Gormley Green  1.122 0.075
      Ryan Green  0.212 0.065
     Cuffe Green  0.793 0.073
  OCaolain    SF -0.098 0.039
```

Discuss: Do small standard errors establish that this dimension measures budget support?

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 59–61.

S05c: Read Wordfish word parameters

Copy and run S05c.

```
# S05c: Inspect word discrimination and baseline frequency
word_id <- match(probe_words, wf$features)
print(data.frame(word = probe_words,
  beta = round(wf$beta[word_id], 3),
  psi = round(wf$psi[word_id], 3)), row.names = FALSE)
```

beta links word use to position. psi represents the word's baseline frequency.

S05c output: discrimination and baseline frequency

R console output

```
       word  beta   psi
     budget 0.042 2.711
 government 0.173 2.612
        tax 0.396 2.194
       jobs 0.505 1.842
       fair 0.191 0.860
   recovery 0.358 0.897
```

Discuss: How does the role of beta differ from both frequency and an ordinary sentiment score?

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 37–38.

S06a: Compare the twelve new texts

Copy and run S06a.

```r
# S06a: Compare rankings among the twelve new texts
comparison <- results[is.na(results$reference), ]
comparison$ws_rank <- rank(comparison$ws_raw)
comparison$wf_rank <- rank(comparison$wf_theta)
comparison$gap <- abs(comparison$ws_rank - comparison$wf_rank)
comparison <- comparison[order(-comparison$gap, comparison$doc_id), ]
round(cor(comparison$ws_raw, comparison$wf_theta,
  method = "spearman"), 3)
print(head(comparison[c("speaker", "ws_rank", "wf_rank", "gap")], 6),
  row.names = FALSE)
```

Exclude both Wordscores references. Rank 1 is the lower end of each scale, after fixing a common orientation.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), p. 59.

S06a output: ranking agreement and disagreements

R console output

```
[1] 0.734
  speaker ws_rank wf_rank gap
   Burton       7       1   6
 ODonnell       2       6   4
    Ryan        6       9   3
 OCaolain      10       7   3
   Cuffe        8      10   2
  Morgan        9       8   1
```

Discuss: Why can overall rank agreement coexist with a large disagreement about one speech?

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), p. 59.

S06b: Plot the two scales

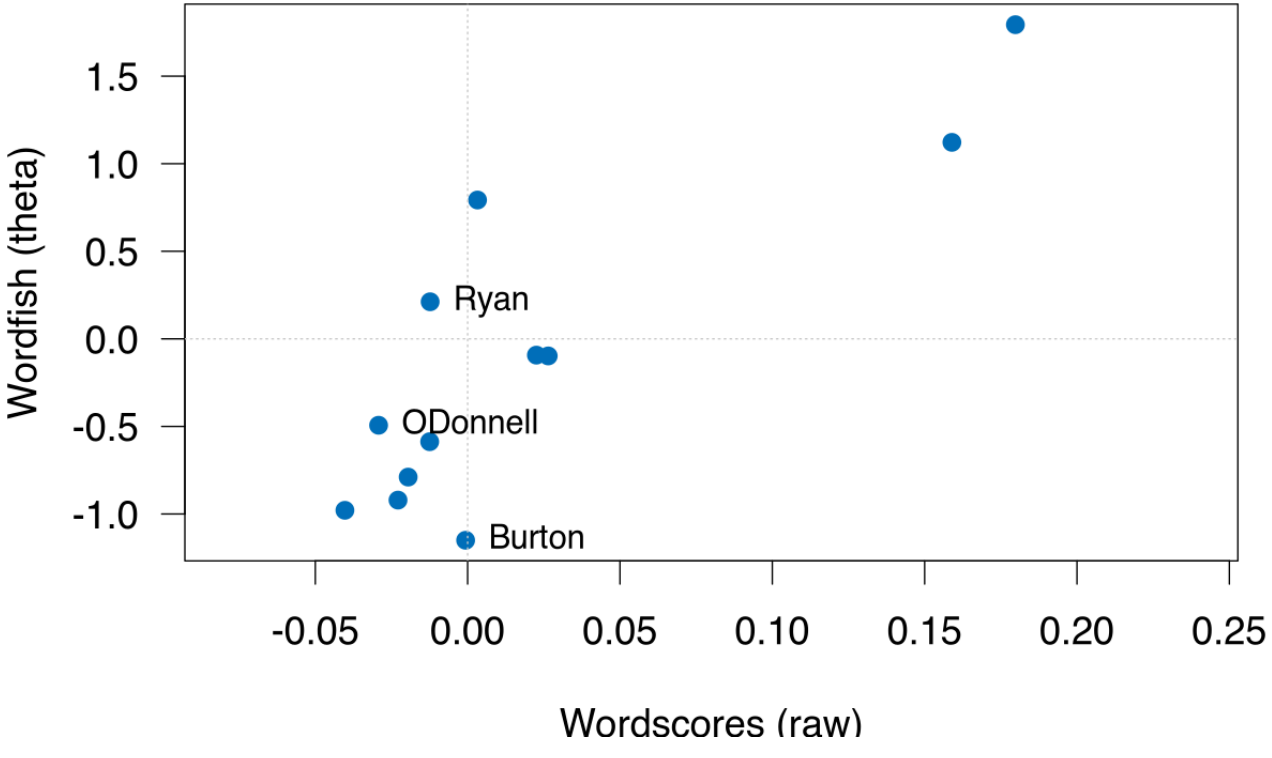
Copy and run S06b.

```
# S06b: Plot the new texts on both scales
plot(comparison$ws_raw, comparison$wf_theta,
  pch = 19, col = "#0271bb", xlim = c(-0.08, 0.24),
  xlab = "Wordscores (raw)", ylab = "Wordfish (theta)")
abline(h = 0, v = 0, col = "grey80", lty = 3)
text(comparison$ws_raw[1:3], comparison$wf_theta[1:3],
  labels = comparison$speaker[1:3], pos = 4, cex = 0.85)
```

Each point is one of the twelve new texts. Labels identify the three largest ranking disagreements.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), p. 41.

S06b output: two scales for the new texts



Discuss: Can the numerical distance on one axis be read in the units of the other?

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), p. 41.

S07a: Read a ranking disagreement

Copy and run S07a.

```r
# S07a: Identify and read the largest ranking disagreement
read_id <- comparison$doc_id[1]
print(comparison[1, c("doc_id", "speaker", "ws_rank", "wf_rank")],
  row.names = FALSE)
read_sentences <- as.character(corpus_reshape(
  corp_irish[read_id], to = "sentences"))
cat(strwrap(read_sentences[1], width = 66), sep = "\n")
```

Use the largest ranking gap to select a speech, then return to its original wording.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 59, 64–66.

S07a output: the selected speech and its opening

R console output

```
 doc_id speaker ws_rank wf_rank
  ie_03  Burton       7       1
The other day, Goldman Sachs circulated a note about the Irish
budget which referred to the remarkable degree of consensus in
Ireland about the necessity for budget adjustments.
```

Discuss: Who supplied the opening claim about consensus, and how might the speaker use it?

S07b: Read the surrounding argument

Copy and run S07b.

```
# S07b: Read more of the same speech
# In this corpus, the selected speech is Burton's. Read sentences 5–7.
cat(strwrap(paste(read_sentences[5:7], collapse = " "),
  width = 66), sep = "\n")
# For the full speech in your console:
# cat(as.character(corp_irish[read_id]))
```

Read sentences 5–7 in Burton’s speech. The original corpus preserves their punctuation and wording.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 64–66.

S07b output: more of the same speech

R console output

```
One could never have imagined one's party giving such an
advantageous start to any Government. This could have been a
consensus budget as it was a classic "all jump together" moment,
an opportunity for burden sharing by everyone but the Minister
blew it. The budget is straight out of the Leona Helmsley school
of thought in which burden sharing, like taxation, is only for
the little people.
```

Discuss: Does this passage support interpreting a middle score as moderate budget support? What else would you read?

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 64–66.

S08a: Change one reference text

Copy and run S08a.

```r
# S08a: Replace the positive reference and refit Wordscores
ref_alt <- ref_scores
ref_alt[c("ie_01", "ie_05")] <- c(NA, 1)
ws_alt <- textmodel_wordscores(counts, y = ref_alt, smooth = 0)
ws_alt_counts <- dfm_match(counts, features = names(coef(ws_alt)))
ws_alt_pred <- predict(ws_alt, newdata = ws_alt_counts,
  rescaling = "none")
print(data.frame(speaker = meta$speaker[!is.na(ref_alt)],
  reference = ref_alt[!is.na(ref_alt)]), row.names = FALSE)
```

Replace Lenihan with Cowen at +1. Keep Bruton at −1 and use the same preprocessing and raw scoring rule.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 17–20.

S08a output: the alternative reference design

R console output

```
  speaker reference
   Bruton        −1
    Cowen         1
```

Discuss: Which part of the measurement design changed, even though the assigned endpoints stayed at −1 and +1?

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 17–20.

S08b: Compare a common set of new texts

Copy and run S08b.

```
# S08b: Compare ranks within the same eleven new texts
common_new <- is.na(ref_scores) & is.na(ref_alt)
sensitivity <- meta[common_new, c("doc_id", "speaker")]
sensitivity$before <- rank(ws_pred[common_new])
sensitivity$after <- rank(ws_alt_pred[common_new])
sensitivity$change <- sensitivity$after - sensitivity$before
print(sensitivity[c("speaker", "before", "after", "change")],
  row.names = FALSE)
```

Recalculate both rankings within the same eleven texts. Exclude Lenihan, Bruton and Cowen from this comparison.

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 17–20.

S08b output: sensitivity to the positive reference

R console output

```
   speaker before after change
    Burton      7     5     -2
    Morgan      9     8     -1
     Kenny      4     2     -2
  ODonnell      2     4      2
   Gilmore      5     6      1
   Higgins      1     1      0
     Quinn      3     3      0
   Gormley     11    11      0
      Ryan      6     9      3
     Cuffe      8    10      2
  OCaolain     10     7     -3
```

Discuss: Which rankings change most, and how should that affect the strength of the substantive claim?

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), pp. 17–20.

Reporting a text-scaling analysis

Decision	What to report
Corpus and representation	Debate, speakers, preprocessing and counts
Wordscores	Reference texts, assigned values, coverage and raw/rescaled scores
Wordfish	Direction, model settings and the interpretation of theta
Assessment	Ranking disagreements, passages read and sensitivity

Return to Lowe & Benoit: what independent reader judgments would test our budget-support interpretation?

W7 run: Irish 2010 budget corpus (`quanteda.textmodels`). Lab HTML · Lab ZIP. Concepts: Benoit (2018), p. 59.

References: text scaling foundations

Benoit, Kenneth. 2018. *Scaling Text*. Quants 3: Quantitative Text Analysis, Trinity College Dublin, March 23. Lecture reference for the concept walkthrough, pp. 3–27, 34–50, 54–55, 59–70.

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Slapin, Jonathan B., and Sven-Oliver Proksch. 2008. "A Scaling Model for Estimating Time-Series Party Positions from Texts." *American Journal of Political Science* 52(3): 705–722. DOI.

Lauderdale, Benjamin E., and Alexander Herzog. 2016. "Measuring Political Positions from Legislative Speech." *Political Analysis* 24(3): 374–394. DOI.

References: validation and semantic scaling

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Lowe, Will, and Kenneth Benoit. 2013. "Validating Estimates of Latent Traits from Textual Data Using Human Judgment as a Benchmark." *Political Analysis* 21(3): 298–313. DOI.

Watanabe, Kohei. 2021. "Latent Semantic Scaling: A Semisupervised Text Analysis Technique for New Domains and Languages." *Communication Methods and Measures* 15(2): 81–102. DOI.

Trubowitz, Peter, and Kohei Watanabe. 2021. "The Geopolitical Threat Index: A Text-Based Computational Approach to Identifying Foreign Threats." *International Studies Quarterly* 65(3): 852–865. DOI.

Software and source materials

- quanteda and quanteda.textmodels: corpus preparation, Wordscores and Wordfish.
- wordshoal: cross-debate scaling and implementation documentation.
- LSX: semantic scaling and seed-word evaluation.
- Quanteda scaling tutorials: worked model examples.

Article images reproduce the original research evidence. The participation table and numerical Wordscores example are teaching illustrations. Speech excerpts retain their original wording.

Image provenance and source-page coordinates: sources.json. Reading excerpts: provenance. Benoit lecture: PDF · Page guide.